Title : Implement a Linear Regression Model to predict house prices for regions in the USA using the provided dataset.

1) Import Required Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn a s sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

sns.set_style('whitegrid')
np.random.seed(42)
```

2) Load the Dataset

```
data_url = 'https://raw.githubusercontent.com/huzaifsayed/Linear-Regression-Model-for-House-Price-Prediction/master/USA_Housing.
df = pd.read_csv(data_url)

print('Dataset shape:', df.shape)
df.head()
```

```
Dataset shape: (5000, 7)
```

	Avg. Area Income	Avg. Area House Age	Avg. Area Number of Rooms	Avg. Area Number of Bedrooms	Area Population	Price	Address
0	79545.458574	5.682861	7.009188	4.09	23086.800503	1.059034e+06	208 Michael Ferry Apt. 674\nLaurabury, NE 3701...
1	79248.642455	6.002900	6.730821	3.09	40173.072174	1.505891e+06	188 Johnson Views Suite 079\nLake Kathleen, CA...
2	61287.067179	5.865890	8.512727	5.13	36882.159400	1.058988e+06	9127 Elizabeth Stravenue\nDanieltown, WI

3) Data Understanding and Pre-processing

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 7 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   Avg. Area Income             5000 non-null   float64
 1   Avg. Area House Age          5000 non-null   float64
 2   Avg. Area Number of Rooms    5000 non-null   float64
 3   Avg. Area Number of Bedrooms 5000 non-null   float64
 4   Area Population              5000 non-null   float64
 5   Price                        5000 non-null   float64
 6   Address                      5000 non-null   object
dtypes: float64(6), object(1)
memory usage: 273.6+ KB
```

```
df.describe()
```

	Avg. Area Income	Avg. Area House Age	Avg. Area Number of Rooms	Avg. Area Number of Bedrooms	Area Population	Price
count	5000.000000	5000.000000	5000.000000	5000.000000	5000.000000	5.000000e+03
mean	68583.108984	5.977222	6.987792	3.981330	36163.516039	1.232073e+06
std	10657.991214	0.991456	1.005833	1.234137	9925.650114	3.531176e+05
min	17796.631190	2.644304	3.236194	2.000000	172.610686	1.593866e+04
25%	61480.562388	5.322283	6.299250	3.140000	29403.928702	9.975771e+05
50%	68804.286404	5.970429	7.002902	4.050000	36199.406689	1.232669e+06
75%	75783.338666	6.650808	7.665871	4.490000	42861.290769	1.471210e+06
max	107701.748378	9.519088	10.759588	6.500000	69621.713378	2.469066e+06

```
# Check missing values
df.isnull().sum()
```

	0
Avg. Area Income	0
Avg. Area House Age	0
Avg. Area Number of Rooms	0
Avg. Area Number of Bedrooms	0
Area Population	0
Price	0
Address	0

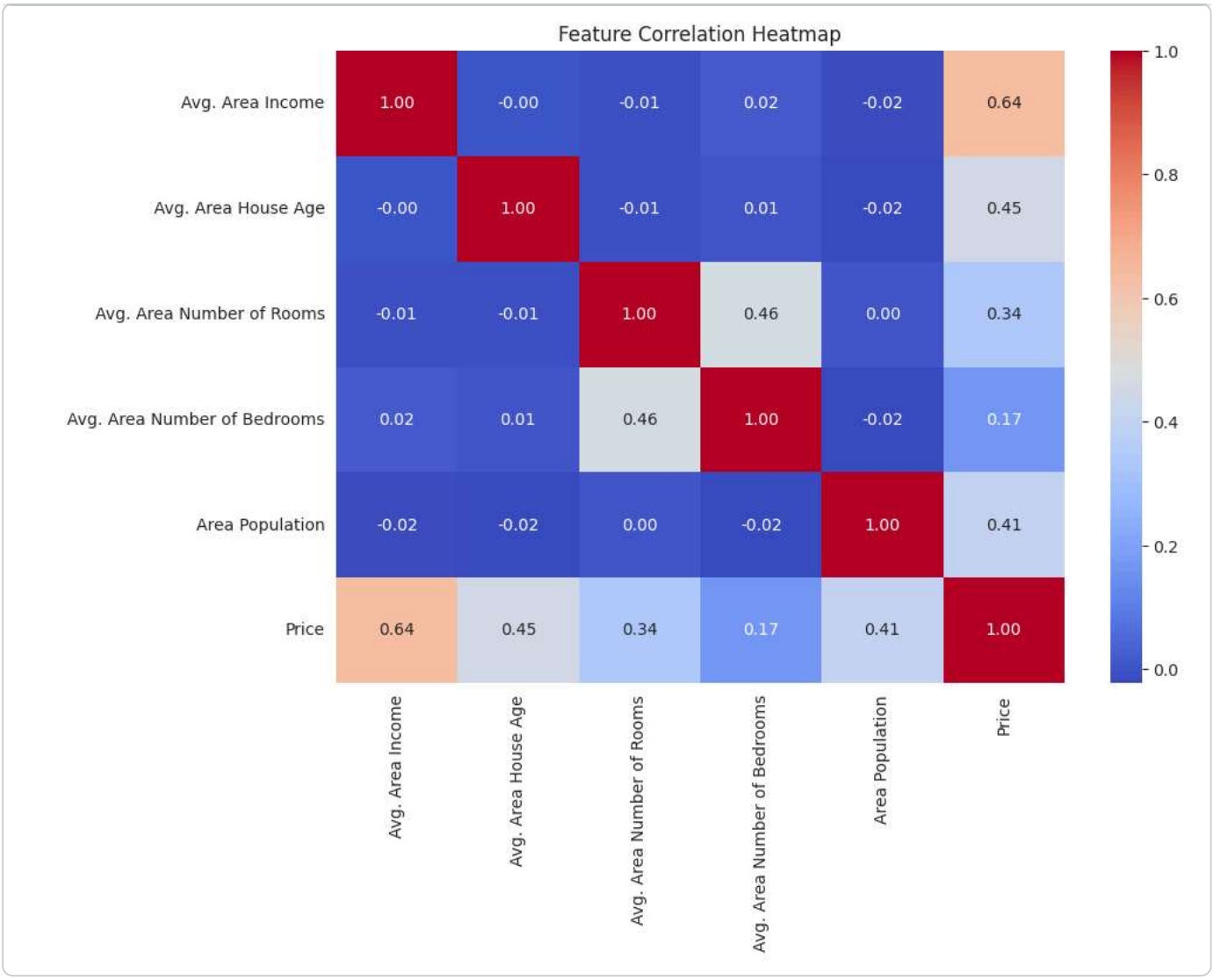
dtype: int64

```
# Drop non-numeric identifier column (Address) before modeling
df_model = df.drop('Address', axis=1)
df_model.head()
```

	Avg. Area Income	Avg. Area House Age	Avg. Area Number of Rooms	Avg. Area Number of Bedrooms	Area Population	Price
0	79545.458574	5.682861	7.009188	4.09	23086.800503	1.059034e+06
1	79248.642455	6.002900	6.730821	3.09	40173.072174	1.505891e+06
2	61287.067179	5.865890	8.512727	5.13	36882.159400	1.058988e+06
3	63345.240046	7.188236	5.586729	3.26	34310.242831	1.260617e+06
4	59982.197226	5.040555	7.839388	4.23	26354.109472	6.309435e+05

4) Exploratory Data Analysis (EDA)

```
plt.figure(figsize=(10, 7))
corr = df_model.corr(numeric_only=True)
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Feature Correlation Heatmap')
plt.show()
```



5) Define Features and Target

```
X = df_model.drop('Price', axis=1)
y = df_model['Price']

print('Feature matrix shape:', X.shape)
print('Target vector shape :', y.shape)
X.head()
```

```
Feature matrix shape: (5000, 5)
Target vector shape : (5000,)
```

	Avg. Area Income	Avg. Area House Age	Avg. Area Number of Rooms	Avg. Area Number of Bedrooms	Area Population
0	79545.458574	5.682861	7.009188	4.09	23086.800503
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3	63345.240046	7.188236	5.586729	3.26	34310.242831
4	59982.197226	5.040555	7.839388	4.23	26354.109472

6) Train-Test Split

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

print('X_train:', X_train.shape)
print('X_test :', X_test.shape)
```

```
print('y_train:', y_train.shape)
print('y_test :', y_test.shape)
```

```
X_train: (4000, 5)
X_test : (1000, 5)
y_train: (4000,)
y_test : (1000,)
```

7) Train Linear Regression Model

```
lr_model = LinearRegression()
lr_model.fit(X_train, y_train)

print('Model training completed.')
```

```
Model training completed.
```

8) Model Coefficients and Intercept

```
coeff_df = pd.DataFrame({
    'Feature': X.columns,
    'Coefficient': lr_model.coef_
})

print('Intercept:', lr_model.intercept_)
coeff_df
```

```
Intercept: -2635072.900933358
```

	Feature	Coefficient
0	Avg. Area Income	21.652206
1	Avg. Area House Age	164666.480722
2	Avg. Area Number of Rooms	119624.012232
3	Avg. Area Number of Bedrooms	2440.377611
4	Area Population	15.270313

9) Predictions and Evaluation

```
y_pred = lr_model.predict(X_test)

mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print(f'MAE  : {mae:,.2f}')
print(f'MSE  : {mse:,.2f}')
print(f'RMSE : {rmse:,.2f}')
print(f'R2   : {r2:.4f}')
```

```
MAE  : 80,879.10
MSE  : 10,089,009,300.89
RMSE : 100,444.06
R2   : 0.9180
```

```
results_df = pd.DataFrame({
    'Actual Price': y_test,
    'Predicted Price': y_pred
})
results_df.head(10)
```

	Actual Price	Predicted Price
1501	1.339096e+06	1.308588e+06
2586	1.251794e+06	1.237037e+06
2653	1.340095e+06	1.243429e+06
1055	1.431508e+06	1.228900e+06
705	1.042374e+06	1.063321e+06

```
plt.figure(figsize=(7, 6))
plt.scatter(y_test, y_pred, alpha=0.6)
plt.xlabel('Actual Price')
plt.ylabel('Predicted Price')
plt.title('Actual vs Predicted House Prices')

# Ideal line
min_val = min(y_test.min(), y_pred.min())
max_val = max(y_test.max(), y_pred.max())
plt.plot([min_val, max_val], [min_val, max_val], 'r--')

plt.show()
```